

Masked-Media Protocol v3

A receiver-gated, signed and encrypted masked-media protocol for image, audio and video carriers.

- AES-256-GCM
- RSA-2048
- SHA-256
- 1-8 LSB



ENCRYPT → SIGN → EMBED → VERIFY

Security Properties and Trust Model

01



Confidential payload

The complete record—not only the user bytes—is encrypted with AES-256-GCM.

02



Intended receiver

RSA-OAEP hides the packet location, depth, length and AES session material.

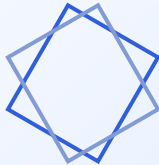
03



Trusted sender

RSA-PSS authenticates the ciphertext, carrier interpretation and embedding geometry.

04

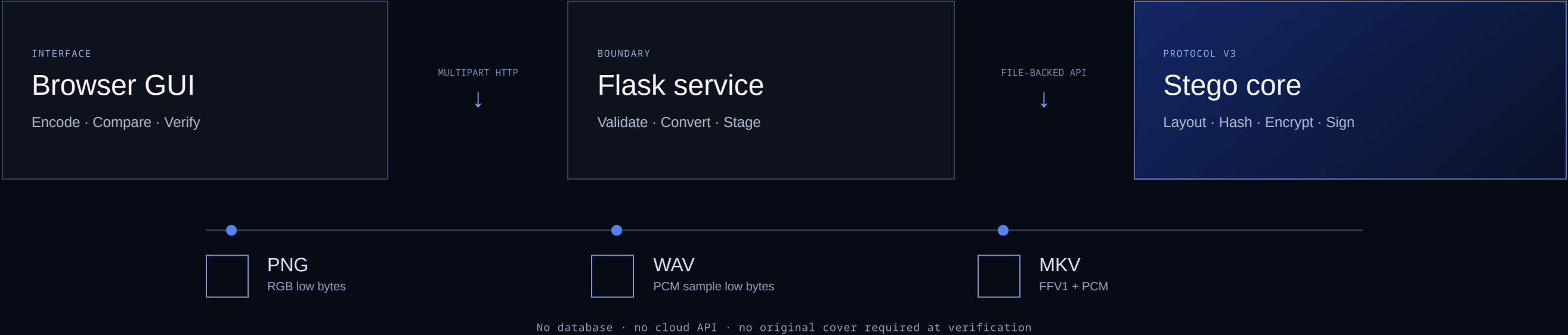


Media integrity

A reproducible masked SHA-256 hash detects changes without needing the original cover.

Payload bytes are released only when every reachable check produces **Authentic**.

Application and Protocol Architecture

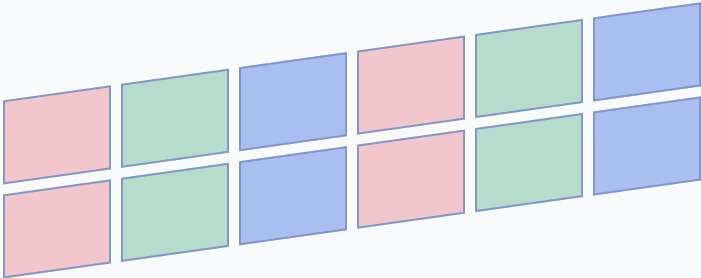


One Protocol, Three Carrier Adapters

01

PNG

IMAGE



Carrier unit: low byte of R, G or B.

Order: row-major, R → G → B.

Fixed data: alpha and 16-bit high bytes.

02

WAV

AUDIO



Carrier unit: low byte of each PCM sample.

Order: declared interleaved samples.

Fixed data: every other sample byte.

03

MKV

OPTIONAL VIDEO



Carrier unit: RGB low bytes, then s16 audio low bytes.

Fixed data: timing, high bytes and alpha.

Output: lossless FFV1 + PCM.

Embedding Data Through LSB Replacement

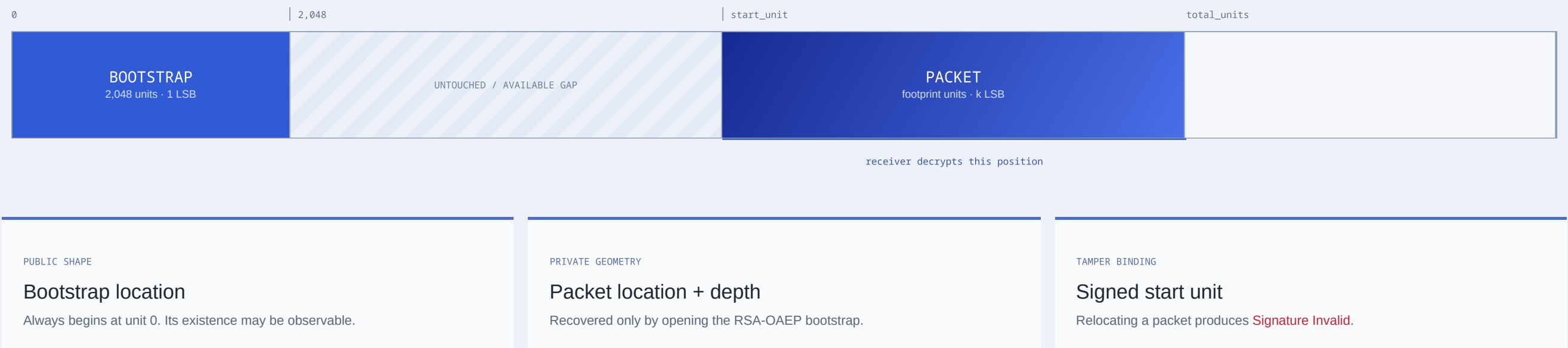


```
stego = (carrier & ~((1 << k) - 1)) | packet_field
```

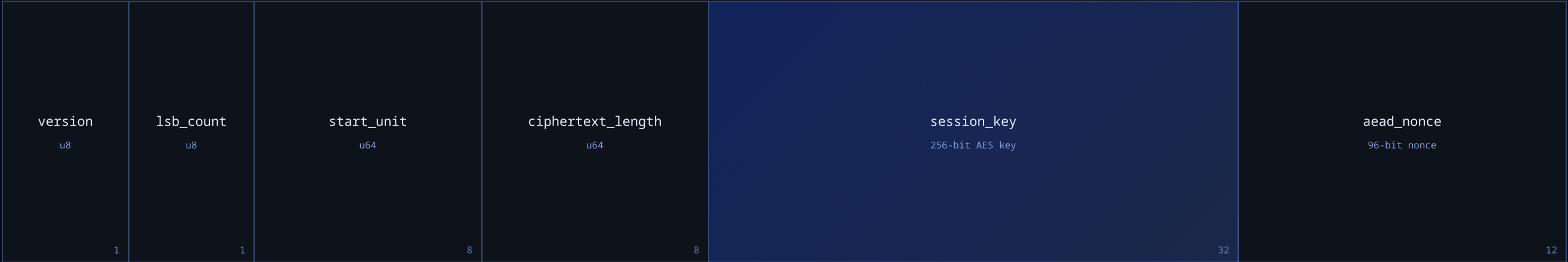
Bytes become bits MSB-first; the packet field occupies the low k bits.



Fixed Bootstrap, Relocatable Packet



Receiver-Only Packet Discovery



AES-GCM AAD / 18 BYTES version || lsb_count || start_unit || ciphertext_length

The geometry recovered from the bootstrap must also authenticate under the payload encryption.

Encrypted Payload Record Layout



Separate Cryptographic Security Roles



Exactly What the Sender Signs

DOMAIN INF2005-ACW1\0SIGN\0	INTERPRETATION version · media_code · k	GEOMETRY total · start · footprint · length	MEDIA CONTEXT dimensions / PCM format / video timing shape	CONTENT complete AES-GCM ciphertext + tag
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Integrity Without the Original Cover



Same logical media + different embedded bits ⇒ same masked hash.
Changed preserved media ⇒ **Tampered**.

LSB Depth Controls Capacity and Distortion

SELECT PACKET LSB DEPTH

1

2

3

4

5

6

7

8

subtle

maximum capacity

EXAMPLE PACKET

448 B

3,584 bits

FOOTPRINT

3,584

carrier units

PACKET UNIT BITS PRESERVED

87.5%

7 of 8 bits

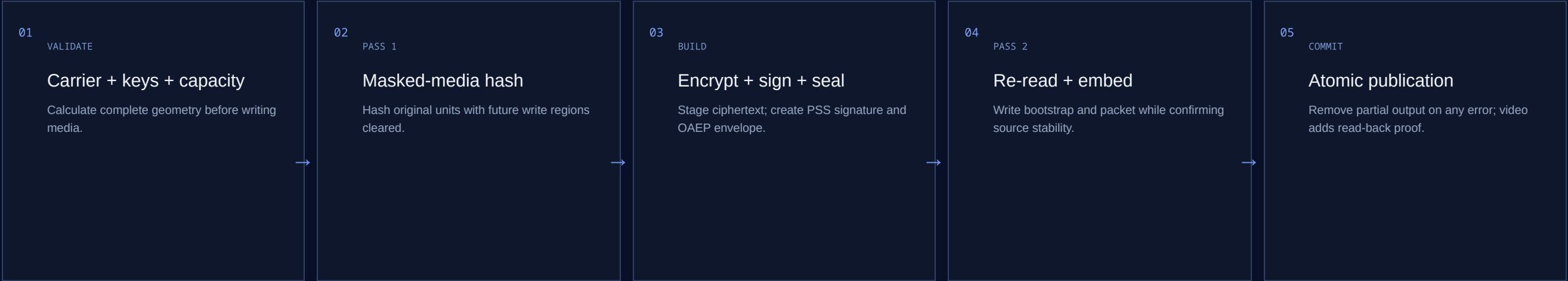
packet footprint

same packet / fewer units as k increases

$$\text{footprint} = \text{ceil}(((\text{ciphertext_length} + 256) \times 8) / k)$$

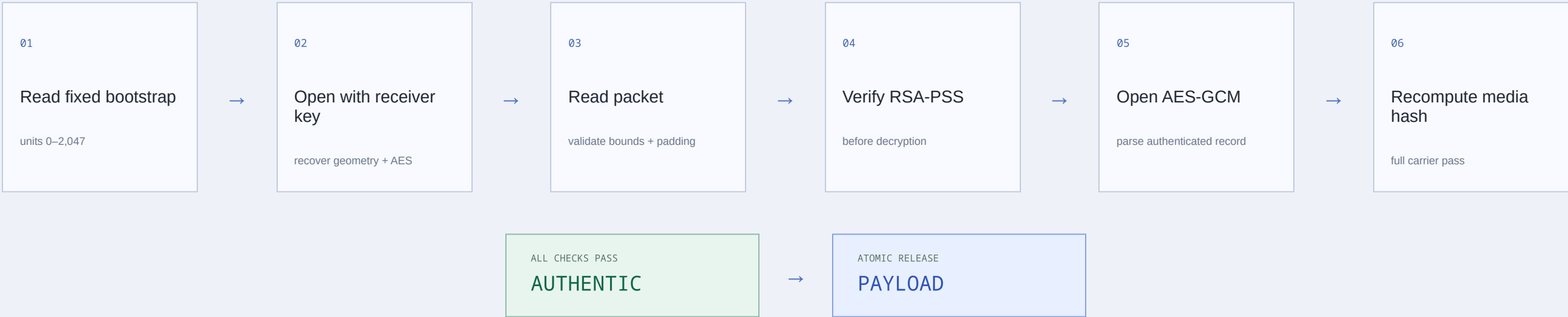
$$\text{max payload} = \text{floor}((\text{total_units} - \text{start_unit}) \times k / 8) - \text{overhead}$$

Validated Two-Pass Encoding



● Source changed between passes? **ABORT** Output differs after video mux? **ABORT**

Proof Before Plaintext Release

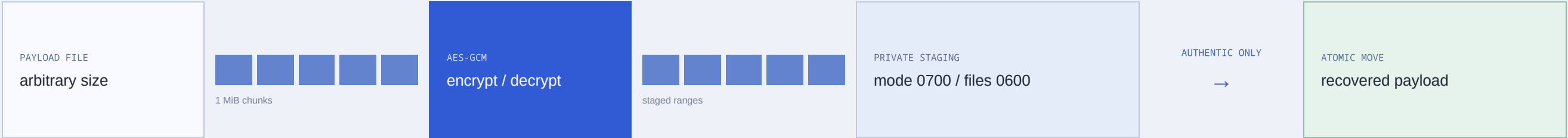


Receiver provides only: stego media · trusted sender public key · receiver private key

Defined Outcomes for Every Failure Stage

ALL GATES Authentic Signature, GCM and media hash agree.	RECEIVER_GATE Payload Missing Bootstrap cannot be opened.	GEOMETRY Wrong Start Location Packet overlaps or exceeds carrier.	SENDER_PROOF Signature Invalid PSS rejects signed bytes or layout.
	RECORD_AUTH Cannot Decrypt GCM tag rejects signed ciphertext.	MEDIA_INTEGRITY Tampered Signed record valid; media hash differs.	FORMAT_READ Cannot Verify Malformed, unsupported or incomplete.

Bounded-Memory File Processing



3 GiB reserve

Upload and output operations keep shared free-space headroom.

Failure-safe

No partial carrier or unauthenticated payload is published.

Crash caveat

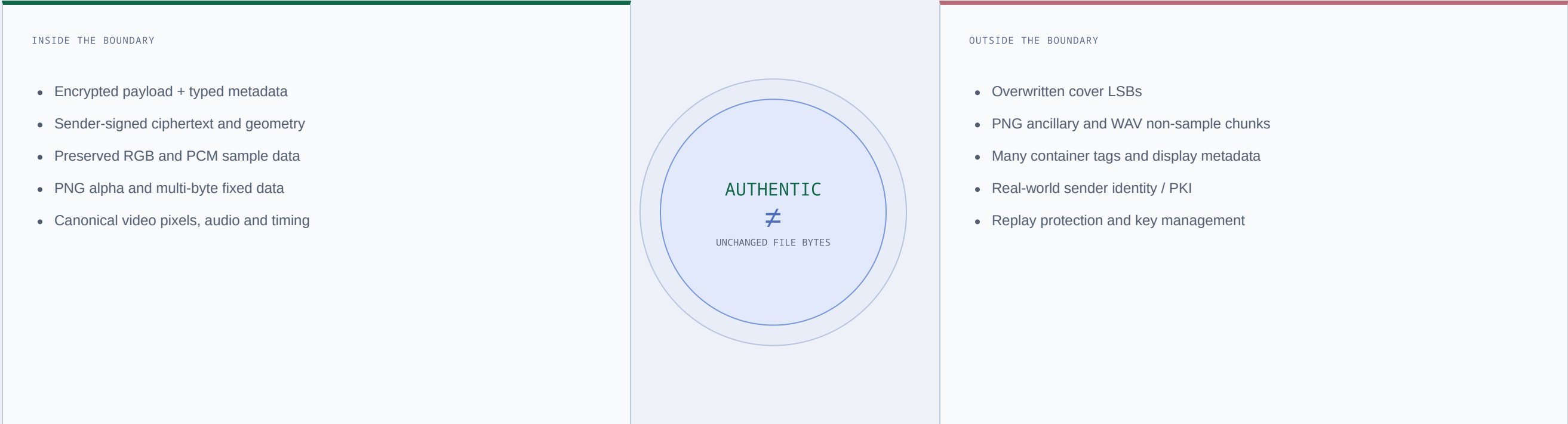
SIGKILL or power loss can leave private staging for manual cleanup.

Canonical Carriers From Common Media Sources

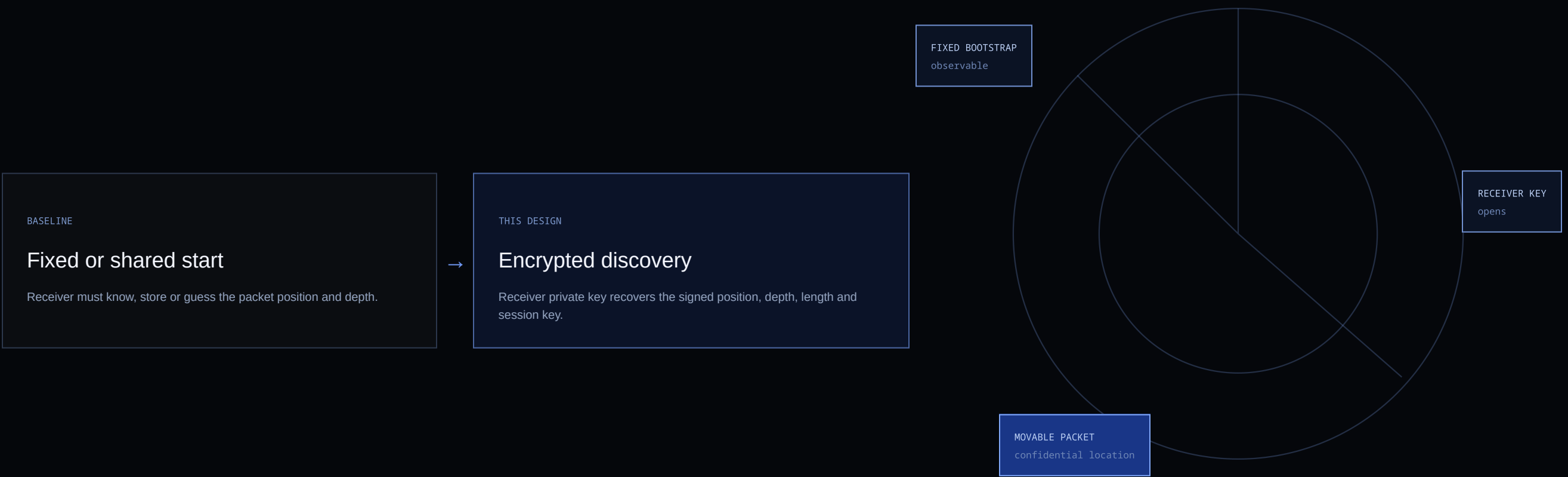


The protocol authenticates canonical decoded media—not the original compressed source bytes or encoder settings.

What an Authentic Verdict Proves



Encrypted Bootstrap Protects Packet Geometry



Evidence From Tests and Demonstration

309

Python tests passed

4

browser-boundary tests passed

1-8

LSB depths round-trip

7

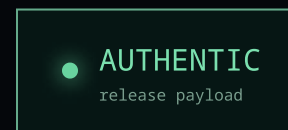
defined verification verdicts

CASE	PNG	WAV	EXPECTED
Correct keys + unchanged media	✓	✓	Authentic
Preserved media bit changed	✓	✓	Tampered
Wrong sender public key	✓	✓	Signature Invalid
Wrong receiver private key	✓	✓	Payload Missing
Payload exceeds capacity	✓	✓	Refused before write

PROTOCOL V3 / TECHNICAL DESIGN

Protocol Verified. Payload Released.

receiver access + sender authenticity + media integrity



Questions / live demonstration